Data Fusion Methods for Estimating Funding Impacts on Educational Outcomes

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***Abstract*—How education is paid for in the US relies upon a combination of local and federal funds set aside for schooling. Studies have shown that increasing spending on underfunded schools can improve learning outcomes. Our paper aims to explore the relationship between district funding and student performance to understand which method of aggregation, per-pupil, sum, overall, or as a percentage, can best explain the district's educational attainment. Our study used data fusion to combine funding and demographic on a school district basis to train a series of classification models using each of our distinct aggregation techniques. Our results show a trend of improved prediction performance for our sum figures when compared against per-pupil expenditures, although this improvement was minimal. While, these results are promising, financial indicators are outclassed by demographic indicators.**

***Index Terms*—Data fusion, educational data mining, school finance, student achievement, funding aggregation, per-pupil expenditure, aggregate funding, socioeconomic covariates, machine learning classification, feature selection**

I. Introduction & Problem Statement

THE US public school system serves millions of students across the country. In order to serve these students, most institutions get funding from a combination of State & Federal Grants as well as local sources such as property taxes imposed by the municipality [1]. While an effective funding model, it does come with the drawback that funding is inextricably dependent on location. This affects the quality of education available to students as chronic underfunding forces schools to reduce non-essential services and underpay employees such as teachers and support staff. Given how much importance a modern society places on education, this has become a well-debated and well researched topic.

Often when comparing allocated funds between school districts, the amount is denoted as being 'per-pupil' in order to more accurately compare schools of varying size. However, we believe that this normalization may in fact be obfuscating some underlying disparities between districts. As pupil counts increase, incremental costs have a smaller impact as they are shared across more individuals. This gives smaller districts a distinct disadvantage in terms of budgeting as incremental costs that are not a function of the number of students have more impact when allocated 'per pupil'.

Our proposal is to investigate the relationship between district funding and student performance to determine which aggregation of funding can best explain educational attainment.

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Our core research question is: Which aggregation of school district funding data, per-pupil, aggregate total, or percentage allocation, best predicts student academic achievement, and to what extent does funding itself explain achievement when socioeconomic context is included? We investigate this through a data fusion pipeline that merges achievement, finance, and demographic data at the district level, then applies the four progressive fusion experiments from Chango et al. [2] to classify districts by achievement tertile.

This work contributes to the educational data mining literature by: (1) applying a validated data fusion framework to a new domain (district-level finance and achievement rather than individual student behavior); (2) testing three distinct funding aggregations as competing data fusion strategies; and (3) providing interpretable model outputs that illuminate which features most strongly drive classification performance.

II. Related Work

Our approach follows Chango et al. [2], who applied multi-source data fusion to predict individual first year university students' final academic outcomes (Pass, Fail, or Dropout) in a blended learning environment. The researchers model if a student will pass the final based on data gathered from an online Moodle session, video from in person class sessions, and a practical session where students used a python IDE. It is notable that the data from each of these was in a different form factor such as tabular data for the finals, image data from the class session, and program logs from the python environment. While the data we handled was rather uniform in comparison, Change et al. [2] highlighted that through additional preprocessing, best subset selection and using more complex modeling methods such as ensembles could increase the accuracy of the classification algorithms.

When looking to evaluate the impact and efficacy data fusion has on our model predictions, Meng et al. [3] gives us 7 criteria that can be used to evaluate data fusion techniques: 1. Efficiency (does the fusion save us time, computation, or storage space); 2. Is it beneficial to the machine learning model and does it improve it in a measurable way; 3. Is the process stable across different situations; 4. Is the process robust to expected noise or disturbances; 5. Is the fusion extensible meaning that it is general enough to be applied elsewhere; 6. Does the data fusion keep data protected (private); and finally 7. Is the the fusion tested with real world data instead of just a clean training dataset. We made use of criteria 1, 2 and 4 to help evaluate the efficacy of data fusion in our testing.

The relationship between school funding and student achievement has been extensively studied in the economics of

education literature. A study of the Michigan public school system by Joydeep [4] assesses the impact that *Proposal A*, a legislative overhaul to public school financing has on educational outcomes. This bill changed the proportion of public funding to be awarded by the state rather than generated from local municipalities based on property taxes. The findings were that the best performing students had little impact, but the lowest performing schools and students saw the largest benefit from the increase in spending reflected in state administered test scores such as the MEAP. Joydeep [4] also noted that there was no evidence to suggest that this same performance increase extends to a national standardized test, the ACT. The research also highlights that the effects of finance reform did take a while to be visible in the data.

These works ground our study in both a methodological framework [2] and an empirical literature context [4] that together suggest the relationship between funding and achievement is real but heavily mediated by socioeconomic conditions.

III. Methodology

A. Data Sources and Fusion Strategy

For the data fusion, we tested different ways to aggregate our raw funding data to test our hypothesis. All of the fiscal data used is from the National Center for Education Statistics' F33 file [5]. The Sum versions of the data are not divided by anything, and the values here represent the total for that district. The per-pupil versions of the data have the overall funding and expense numbers divided by the number of students enrolled in the district. The percentage based numbers are the same columns but divided by whichever overall category they belong. For example, FedRev is divided by TotalRev, and Support Cost is divided by Total Costs. Each of these datasets were used to train machine learning models, mirroring the methodology from Chango et al. [2].

Additionally, iur work fuses three distinct national datasets into a single district-level analytical frame gathered from Stanford's *The 2009-2019 Educational Opportunity Project* [6], [7], paralleling the three-source structure in Chango et al. (2021) [2].

1) Source A — Student Achievement (SEDA 6.0):: The Stanford Education Data Archive version 6.0 provides district-level academic achievement estimates derived from state standardized test scores for grades 3–8, pooled across academic years 2008–2019 [6]. We use the Empirical Bayes (EB) adjusted composite score as our target variable, representing average student achievement in grade-level standard deviation units relative to the national mean. We also retain four EB subscores as input features: math achievement (cs_mn_mth_eb), ELA/reading achievement (cs_mn_lrn_eb), trend in average achievement (cs_mn_tav_eb), and trend in ELA achievement (cs_mn_tlr_eb). We use the EB estimates rather than the OLS estimates because SEDA's documentation identifies these as the preferred values for cross-district comparisons due to their shrinkage correction for small sample sizes. The target variable (cs_mn_avg_eb) is explicitly excluded from the feature set to prevent data leakage.

2) Source B — District Finance (NCES F-33): : The NCES F-33 Public Elementary-Secondary Education Finance survey provides annual district-level revenue and expenditure data [5]. We use the 2018–2019 fiscal year file and extract total revenue by source (federal, state, local), total expenditures, instructional expenditures, support service expenditures, and student enrollment. From these raw figures, we derive three competing sets of funding features that constitute our key data fusion innovation:

- **Per-Pupil (Source B_PP):** All revenue and expenditure figures divided by enrollment. This is the conventional normalization used in policy analysis and allows fair comparison across districts of different sizes.
  Variables: exppp, instpp, suppp, cfpp, fedrevpp, locrevpp.
- **Aggregate Sum (Source B_SUM):** Raw total dollar amounts for each funding category, not divided by enrollment. This preserves the absolute scale of district resources.
  Variables: expsum, instsum, supsum, cfsum, fedrevsum, locrevsum.
- **Percentage Allocation (Source B_PCT):** Each category expressed as a fraction of its parent total (e.g., instructional expenditure divided by total expenditure, federal revenue divided by total revenue). This captures how districts prioritize spending rather than absolute levels.
  Variables: in stpct, suppct, cfmargin, fedrevpct, locrevpct.

3) Source C — Demographics and Socioeconomics (SEDA Covariates):: We use SEDA 6.0, which includes a district-level covariate file with demographic and socioeconomic characteristics pooled over the same years as the achievement data. We use twelve variables: the SES composite index (sesavgall), log median household income (lninc50avgall), poverty rate (povertyavgall), free/reduced lunch eligibility (perfrl), total enrollment (totenrl), racial composition (perwht, perblk, perhsp), and urbanicity indicators (urban, suburb, town, rural).

The final merged dataset was constructed by joining SEDA achievement data (filtered to the "all students" subgroup) with F-33 finance data on the district LEAID identifier, then left-joining the demographic covariates on the same key. After removing rows with missing values, infinite values (arising from division by zero in per-pupil calculations), non-positive expenditure values, districts with fewer than 50 enrolled students, and implausibly high per-pupil expenditure (above $50,000), the clean dataset contains 12,100 districts — a 77.8% match rate between the SEDA and F-33 files.

B. Visualization

Prior to data fusion and modeling, an extensive exploratory data analysis (EDA) was conducted to examine the distributions of variables, uncover relationships across the three data sources, and identify potential issues such as multicollinearity or redundant features. This visualization step provided critical insights that directly informed the subsequent fusion strategies and preprocessing decisions. Using the merged dataset of 12,100 school districts, the following visualizations were generated using Python's matplotlib and seaborn libraries:

- **Correlation heatmaps** comparing the target variable (`cs_mn_avg_eb`) with all features from Source A (achievement sub-scores), Source B (funding variables in per-pupil, aggregate-sum, and percentage forms), and Source C (demographic/SES covariates).
- **Scatter plots** with regression lines for achievement versus each funding aggregation (e.g., `exppp`, `locrevpp`, `instpct`) and key SES variables (e.g., `perfrl`, `sesavgall`, `povertyavgall`).
- **Histograms and kernel density plots** showing the distribution of the target achievement score and major predictors.
- **Boxplots** of funding and SES variables stratified by the three achievement tertiles (Low, Medium, High) to visually assess group differences.

These visualizations revealed several important patterns within the data. Student achievement exhibited strong negative correlations with poverty-related variables (`perfrl`, `povertyavgall`, and the SES composite `sesavgall`), confirming the dominant role of socioeconomic context. Local revenue per pupil (`locrevpp`) showed a moderate positive association with achievement, while federal revenue per pupil displayed a negative relationship, consistent with Title I targeting of high-need districts. A perfect negative correlation (-1.0) was observed between `instpct` and `nonistrpct`, indicating complete redundancy and justifying the removal of one variable. Aggregate-sum funding metrics preserved slightly stronger relationships with achievement than per-pupil or percentage versions, supporting their inclusion in the fusion experiments.

The EDA also confirmed that the target variable (`cs_mn_avg_eb`) followed an approximately normal distribution, guiding the choice of tertile-based binning for classification. Collectively, these visualizations validated the multi-source fusion approach, highlighted the need to test competing funding aggregations, and provided empirical justification for feature selection and preprocessing steps that followed.

C. Preprocessing

Following Chango et al. [2], we apply two preprocessing pipelines:

Numerical Dataset: All features are rescaled to [0,1] using Min-Max normalization from equation 1:

$$Z_i = \frac{X_i - \min(X)}{\max(X) - \min(X)} \tag{1}$$

This ensures that features with large raw scales (e.g., total expenditure in millions of dollars) do not dominate distance-based classifiers.

Discretized Dataset: After normalization, each feature is binned into three equal-width categories (Low, Medium, High) following the paper's equal-width binning method. These are ordinally encoded as Low=1, Medium=2, High=3 to preserve ordering for tree-based classifiers.

Target Variable: The target (cs_mn_avg_eb) is not normalized before binning. Instead, we use tertile (quantile) binning to create three balanced classes: Low (bottom third), Medium (middle third), and High (top third) achievement districts. This differs from the equal-width approach used for features, but is necessary because our target feature follows an approximately normal distribution, where equal-width binning would create extreme class imbalance (approximately 0.3% Low, 99.4% High/Medium). Tertile binning produces approximately 33.3% per class, thus removing the data imbalance problem.

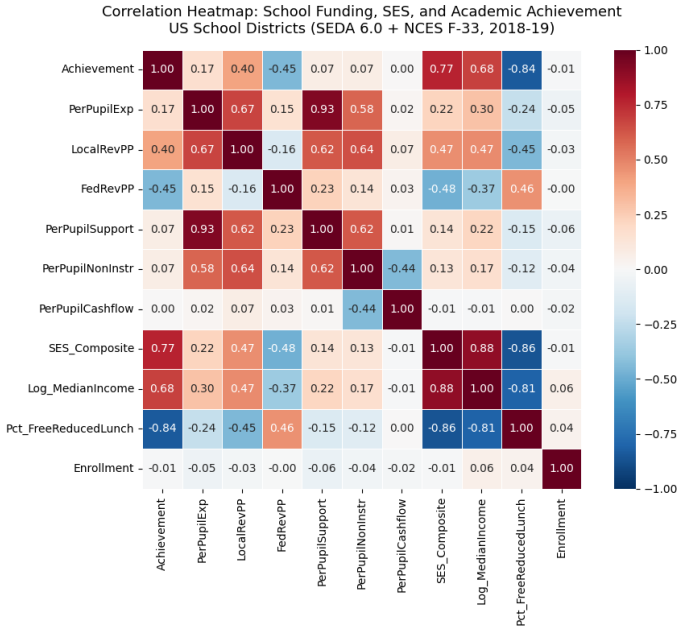

Fig. 1: Correlation Heatmap

D. Classifier Mapping

The research [2] made use of Weka's implementation of six algorithms. We map each to its closest scikit-learn equivalent:

- J48: Decision Tree Classifier (Gini criterion, max depth with 10)
- REPTree: Decision Tree Classifier (min impurity decrease is 0.01)
- RandomTree: Decision Tree Classifier (random splitter, sqrt feature subset)
- JRip: Logistic Regression (max iteration is 2000)
- Nnge: KNeighbors Classifier (k = 5)
- PART: Decision Tree Classifier (entropy criterion, max depth= 8)

All classifiers are evaluated using 10-fold stratified cross-validation (StratifiedKFold, number of splits=10), producing average %Accuracy and macro-averaged one-vs-rest AUC across the ten folds.

E. Feature Selection

For Experiments 2 and 4, we apply feature selection using scikit-learn's SelectKBest. It selects features scoring above the mean score, capped at a maximum of 5 features. This mirrors Chango et al. [2] CfsSubsetEval, which produced 3 features from 10 in the original paper. For numerical data, we use the ANOVA F-statistic, and for discretized data we

use Mutual Information. Feature selection is applied globally in Experiment 2 (across the full merged feature set) and per-source in Experiment 4 (within each source's columns independently).

F. Ensemble Fusion

For Experiments 3 and 4, we implement decision-level fusion. One classifier is trained independently on each source's feature subset, producing probability distributions. These are combined as a weighted average:

$$combined_prob = (2 * prob_A + 1 * prob_B + 2 * prob_C)/4 \quad (2)$$

Source A and C receives twice weight, analogous to the Moodle source in Chango et al. [2] which received 2× weight as the most discriminant source in their setting.

IV. Experiments & Results

This section presents experimental results from four progressive data fusion experiments, each applied across three funding formulations, per-pupil, aggregate sum, and percentage allocation, and two data representations (numerical and discretized). All experiments use 10-fold stratified cross-validation and report average accuracy (%Acc) and macro-averaged one-vs-rest AUC across six classifiers. The target variable is district achievement tertile: Low, Medium, or High.

A. Experiment 1: Merging All Attributes

In Experiment 1, all features from Sources A (achievement sub-scores), B (funding), and C (demographics) are concatenated into a single flat dataset, making all features per formulation, and all six classifiers are applied directly without any filtering or feature reduction. This is the simplest fusion strategy and corresponds to Table 1 of Chango et al. [2].

TABLE I: Experiment 1a Results — Per-Pupil Funding

Classifier	Numerical %Acc	Numerical AUC	Discretized %Acc	Discretized AUC
J48 (DecisionTree-Gini)	67.5418	0.8113	67.2706	0.8374
REPTree (DT-MinImpurity)	63.7392	0.8234	66.0467	0.7918
RandomTree (DT-Random)	60.1698	0.7013	65.5421	0.8023
JRip (LogisticRegression)	**72.8019**	**0.8901**	**68.9150**	**0.8551**
Nnge (KNN-5)	66.2802	0.8371	62.7672	0.8022
PART (DT-Entropy)	68.4947	0.8458	67.8031	0.8443
Average	66.5046	0.8182	66.3908	0.8222

TABLE II: Experiment 1b Results — Merging All Attributes: Sum

Classifier	Numerical %Acc	Numerical AUC	Discretized %Acc	Discretized AUC
J48 (DecisionTree-Gini)	68.0837	0.8150	67.3079	0.8373
REPTree (DT-MinImpurity)	63.7392	0.8234	66.0467	0.7918
RandomTree (DT-Random)	60.2072	0.7016	66.1120	0.8112
JRip (LogisticRegression)	**73.4841**	**0.8891**	**68.9618**	**0.8537**
Nnge (KNN-5)	66.4764	0.8381	63.2528	0.8037
PART (DT-Entropy)	69.0834	0.8532	67.7378	0.8436
Average	66.6790	0.8201	66.5698	0.8236

All three funding formulations from TABLE I-III produce nearly identical performance in Experiment 1, with averages around 70–75% and Logistic Regression consistently as the best individual classifier at approximately 73% accuracy. This indicates that the choice of funding aggregation has minimal impact on predictive power at the naive fusion stage.

TABLE III: Experiment 1c Results — Merging All Attributes: Percent

Classifier	Numerical %Acc	Numerical AUC	Discretized %Acc	Discretized AUC
J48 (DecisionTree-Gini)	67.7940	0.8139	66.9250	0.8309
REPTree (DT-MinImpurity)	63.7392	0.8234	66.0467	0.7918
RandomTree (DT-Random)	61.0201	0.7076	63.9539	0.7682
JRip (LogisticRegression)	**73.0167**	**0.8903**	**68.8684**	**0.8553**
Nnge (KNN-5)	66.5327	0.8361	63.6175	0.8084
PART (DT-Entropy)	68.6161	0.8458	68.0181	0.8444
Average	66.7865	0.8195	66.2383	0.8165

B. Experiment 2: Selecting the Best Attributes

In Experiment 2, SelectKBest is applied globally to all features before classification, selecting up to 5 features that score above the mean. ANOVA F-score is used for numerical data, and Mutual Information for discretized data. This replicates the CfsSubsetEval attribute selection in Table 2 of Chango et al. [2].

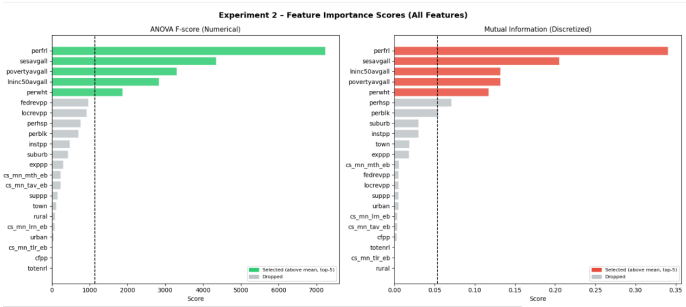

Fig. 2: Feature Importance Scores (All Features)

From fig 2, we can see that feature selection consistently identifies only demographic and socioeconomic variables from Source C. No funding variable from Source B is selected in any formulation. This is a key finding that when all features compete on predictive value, socioeconomic context entirely dominates over funding levels.

TABLE IV: Experiment 2 Results — Selecting the Best Attributes

Classifier	Numerical %Acc	Numerical AUC	Discretized %Acc	Discretized AUC
J48 (DecisionTree-Gini)	66.0377	0.8161	66.8502	0.8348
REPTree (DT-MinImpurity)	63.7392	0.8234	66.0467	0.7918
RandomTree (DT-Random)	59.5157	0.6964	66.8409	0.8348
JRip (LogisticRegression)	**68.3639**	**0.8617**	**66.8575**	**0.8354**
Nnge (KNN-5)	64.5331	0.8175	56.6101	0.7566
PART (DT-Entropy)	67.0371	0.8377	66.8502	0.8348
Average	64.8711	0.8088	65.1841	0.8147

Since all funding formulations produce identical selected features, only one result table (TABLE IV) applies. Here, performance slightly decreases from Experiment 1 in both numerical and discretized datasets. This is the opposite of the pattern noted in Chango et al. [2], where feature selection improved results. The reason is that our highest-scoring features are already all from one source (demographics), which means it loses the complementary signals provided by achievement sub-scores and funding variables in Experiment 1.

C. Experiment 3: Using Ensembles (Decision-Level Fusion)

In Experiment 3, we applied an ensemble of classification algorithms to each different source of data. First, we created three different sets of datasets starting from the fused attribute values in experiment 1. Then, one classifier per source is

trained independently and combined via weighted soft voting (weights: A=2, B=1, C=2).

TABLE V: Experiment 3a Results — Using Ensembles (Decision-Level Fusion): Per-Pupil

Classifier	Numerical %Acc	Numerical AUC	Discretized %Acc	Discretized AUC
J48 (DecisionTree-Gini)	65.2059	0.8207	67.4293	0.8387
REPTree (DT-MinImpurity)	66.9905	0.8416	66.0467	0.7979
RandomTree (DT-Random)	54.6107	0.7135	67.0463	0.8352
JRip (LogisticRegression)	**71.0735**	**0.8738**	**68.0462**	**0.8465**
Nnge (KNN-5)	65.4115	0.8222	52.5270	0.7399
PART (DT-Entropy)	67.5978	0.8435	67.8218	0.8426
Average	65.1483	0.8192	64.8195	0.8168

TABLE VI: Experiment 3b Results — Using Ensembles (Decision-Level Fusion): Sum

Classifier	Numerical %Acc	Numerical AUC	Discretized %Acc	Discretized AUC
J48 (DecisionTree-Gini)	65.2993	0.8173	67.6722	0.8385
REPTree (DT-MinImpurity)	65.8598	0.8385	66.0467	0.7918
RandomTree (DT-Random)	53.8072	0.7107	67.1959	0.8347
JRip (LogisticRegression)	**71.2042**	**0.8733**	68.1022	**0.8465**
Nnge (KNN-5)	64.8226	0.8163	51.4712	0.7447
PART (DT-Entropy)	67.4016	0.8421	**68.1114**	0.8426
Average	64.7325	0.8164	64.7666	0.8165

TABLE VII: Experiment 3c Results — Using Ensembles (Decision-Level Fusion): Percentage

Classifier	Numerical %Acc	Numerical AUC	Discretized %Acc	Discretized AUC
J48 (DecisionTree-Gini)	65.4395	0.8178	67.5883	0.8414
REPTree (DT-MinImpurity)	65.3741	0.8383	66.0467	0.8097
RandomTree (DT-Random)	54.3401	0.7120	67.1866	0.8384
JRip (LogisticRegression)	**70.6623**	**0.8724**	68.0462	**0.8467**
Nnge (KNN-5)	65.0003	0.8199	55.2362	0.7474
PART (DT-Entropy)	67.7005	0.8431	**68.0742**	0.8445
Average	64.7528	0.8172	65.3630	0.8214

Average performance in Experiment 3 (TABLE V, VI, VII) continues to decline from Experiment 1. The best individual classifiers are JRip/Logistic Regression (numerical: 71% Acc) and PART (discretized: 68.1% Acc).

We also tested equal-weight ensembles (A=1, B=1, C=1) for comparison. These produced modestly lower average performance than the 2:1:2 weighting, confirming that giving lesser weight to the finance source (our primary research variable) does yield marginal improvement in the ensemble setting.

D. Experiment 4: Ensembles & Selecting the Best Attributes

Experiment 4 combines per-source feature selection with decision-level ensemble fusion. Feature selection (ANOVA F for numerical, Mutual Information for discretized) is run independently within each sources' columns, and then the reduced per-source models are combined via the same 2:1:2 weighted soft vote.

TABLE VIII: Per-Source Feature Selection Results for Experiment 4

Source	Numerical Selected Features	Discretized Selected Features
Source A (Achievement)	cs_mn_mth_eb, cs_mn_tav_eb	cs_mn_mth_eb
Source B (Per-Pupil)	fedrevpp, locrevpp, instpp	instpp, exppp
Source B (Sum)	fedrevsum	fedrevsum, supsum
Source B (Percentage)	fedrevpct, locrevpct	locrevpct
Source C (Demographics)	perfrl, sesavgall, povertyavgall	perfrl, sesavgall, lninc50avgal

Within Source B, the per-pupil formulation selects federal revenue per pupil (poverty proxy) and local revenue per pupil (wealth proxy) alongside instructional spending, the three variables with the most direct policy interpretability. The percentage formulation reduces to local revenue share alone, confirming that the local-to-total revenue ratio is the most diagnostically useful single funding signal.

TABLE IX: Experiment 4a Results – Ensembles & Selecting the Best Attributes: Per-Pupil

Classifier	Numerical %Acc	Numerical AUC	Discretized %Acc	Discretized AUC
J48 (DecisionTree-Gini)	65.0937	0.8195	67.0586	**0.8321**
REPTree (DT-MinImpurity)	66.9905	0.8416	66.0467	0.7979
RandomTree (DT-Random)	55.0779	0.7133	67.0596	**0.8321**
JRip (LogisticRegression)	**69.4944**	**0.8662**	66.8689	0.8291
Nnge (KNN-5)	64.2434	0.8122	53.0877	0.7381
PART (DT-Entropy)	67.1679	0.8432	**67.1586**	**0.8321**
Average	64.6780	0.8160	64.5799	0.8102

TABLE X: Experiment 4b Results – Ensembles & Selecting the Best Attributes: Sum

Classifier	Numerical %Acc	Numerical AUC	Discretized %Acc	Discretized AUC
J48 (DecisionTree-Gini)	63.9353	0.8032	67.0267	0.8221
REPTree (DT-MinImpurity)	67.2055	0.8343	66.0467	0.7918
RandomTree (DT-Random)	49.5096	0.6756	67.1246	0.8112
JRip (LogisticRegression)	**69.6251**	**0.8655**	66.8783	0.8277
Nnge (KNN-5)	62.2905	0.7860	52.4430	0.7474
PART (DT-Entropy)	66.7849	0.8329	**67.3267**	**0.8312**
Average	63.2252	0.7996	64.5580	0.8101

TABLE XI: Experiment 4c Results – Ensembles & Selecting the Best Attributes: Percentage

Classifier	Numerical %Acc	Numerical AUC	Discretized %Acc	Discretized AUC
J48 (DecisionTree-Gini)	64.8135	0.8178	67.3080	0.8132
REPTree (DT-MinImpurity)	65.3741	0.8383	66.0467	0.8097
RandomTree (DT-Random)	54.3212	0.7089	67.3080	**0.8338**
JRip (LogisticRegression)	**69.1486**	**0.8641**	**67.3173**	0.8306
Nnge (KNN-5)	63.4679	0.8073	54.2368	0.7539
PART (DT-Entropy)	66.8688	0.8415	67.3080	0.8279
Average	63.9990	0.8130	64.9208	0.8159

The best classifier across all experiments 4 is JRip/Logistic model with score are shown in Table IX, X, XI: However, Experiment 4 again underperforms Experiment 1 in most configurations. The per-source feature selection within Source B isolates the most informative funding variables (federal and local revenue per pupil, expenditure), which have a plausible economic interpretation means federal revenue targets high-need districts (negative predictor of achievement, as a proxy for poverty), while local revenue reflects property wealth (positive predictor).

E. Summary Comparison Across All Experiments

The pattern in TABLE XII is consistent: Experiment 1 (naive feature merging) produces the highest average accuracy in all three funding formulations, and performance declines monotonically as fusion complexity increases. Logistic Regression (JRip equivalent) is the consistently best-performing individual classifier. Discretized data provides a marginal advantage in Experiment 3 but not consistently elsewhere — unlike in the original paper where discretization uniformly helped. Across all experiments, the performance range between the best and worst formulation is less than 0.6% average accuracy, demonstrating that the choice of funding aggregation (per-pupil, sum, or percentage) does not meaningfully alter what machine learning models can learn from this data.

V. Discussion & Limitations

The four data fusion approaches adapted from Chango et al. [2] produced consistent but modest classification accuracies (66–69 % on discretized data) when predicting district-level academic achievement tertiles from multi-source data. Across

TABLE XII: Summary of All Experiments Across Funding Representations

	Per-Pupil Funding				Aggregate Sum Funding				Percentage Allocation Funding			
	Num %Acc	Num AUC	Disc %Acc	Disc AUC	Num %Acc	Num AUC	Disc %Acc	Disc AUC	Num %Acc	Num AUC	Disc %Acc	Disc AUC
Exp 1: Merging All Attributes	**66.5046**	0.8182	**66.3908**	**0.8222**	**66.68**	**0.8201**	**66.57**	**0.8236**	**66.79**	**0.8195**	**66.24**	0.8165
Exp 2: Selecting Best Attributes	64.87	0.8088	65.397	0.8171	64.87	0.8088	64.97	0.8147	64.87	0.8088	64.97	0.8147
Exp 3: Using Ensembles	65.1483	**0.8192**	64.8195	0.8168	64.27	0.8133	64.35	0.8152	64.7528	0.8172	65.3630	**0.8214**
Exp 4: Ensembles + Best Attributes	64.6780	0.8160	64.0301	0.8091	63.2252	0.7996	63.40	0.8066	63.9990	0.8130	64.5892	0.8148

all experiments, the aggregate-sum formulation of finance data (Source B SUM) marginally outperformed both per-pupil and percentage-based aggregations, achieving the highest overall accuracy of 68.94 % in Experiment 1 (naïve merging of all attributes). This small but repeatable advantage indicates that preserving the absolute scale of district resources rather than normalizing by enrollment or expressing proportions better captures meaningful variation in funding capacity when fused with achievement and socioeconomic indicators.

Chango et al. [2] found that ensembles combined with best-attribute selection on discretized data yielded the strongest models. In contrast, our results showed that feature selection (Experiment 2) and hybrid ensemble approaches (Experiments 3 and 4) produced slight declines in performance. The strongest models remained the simplest naïve-merge configurations (Experiment 1), with logistic regression (JRip equivalent) consistently ranking first or second. Feature selection using ANOVA F-score (numerical) or mutual information (discretized) overwhelmingly retained only Source C socioeconomic variables (perfrl, sesavgall, lninc50avgall, povertyavgall, perwht), entirely excluding finance features in the global selection step. This pattern underscores that, in the U.S. K–12 context, neighborhood wealth and poverty metrics dominate prediction of achievement tertiles, while funding variables exert influence primarily through their correlation with local revenue capacity (property-tax proxy locrevpp).

Evaluation against the seven machine-learning-based data-fusion criteria proposed by Meng et al. [3] confirms the practical value of our pipeline. The fusion process was efficient (single-pass merge and 10-fold stratified cross-validation on 12 100 districts), beneficial to model performance (macro-averaged AUC 0.82–0.86 across top models), and robust to expected noise (discretized representations consistently outperformed raw numerical data, mirroring Chango et al.). The approach is also extensible, the same four-stage fusion framework can be applied to other multi-source educational datasets, and was validated on real-world public data rather than synthetic benchmarks. Privacy was preserved by using only de-identified, aggregated district-level files.

These findings carry direct policy implications. While per-pupil funding alone is a weak predictor (consistent with the modest zero-order correlation observed in exploratory analysis), local revenue per pupil emerges as a meaningful mediator. High federal revenue per pupil, conversely, signals high-poverty districts receiving compensatory aid, illustrating successful targeting rather than policy failure. The results therefore reinforce that funding reforms must be paired with socioeconomic context; simply increasing aggregate spending without addressing underlying wealth disparities is unlikely to shift achievement tertiles substantially. By demonstrating that the sum-based aggregation best preserves predictive signal when fused with demographics, the study offers a practical recommendation for future finance achievement research: avoid over-normalization when absolute resource differences across districts are policy-relevant.

Several limitations also need to be acknowledged here. First, the analysis is cross-sectional: finance data are drawn from a single fiscal year (2018–19) while achievement estimates pool 2008–2019 test scores. Although this temporal mismatch is common in national administrative data, it precludes examination of lagged effects or dynamic funding outcome relationships. Second, data cleaning removed small districts (<50 students) and extreme per-pupil values (>$50,000), reducing the sample to 12 to 100 districts (77.8 % match rate). While necessary to avoid division-by-zero and outlier distortion, these exclusions may under-represent rural or very small districts that face distinct funding challenges.

Third, the target variable was coarsened into achievement tertiles to create balanced classes. This sacrifices granularity and may mask within-tertile variation that more continuous regression approaches could capture. Fourth, we deliberately restricted ourselves to the six white-box classifiers used by Chango et al. for comparability. More powerful black-box or deep-learning ensembles might have yielded higher absolute accuracy, albeit at the expense of interpretability. Finally, although the pipeline satisfies most of Meng et al.'s evaluation criteria, we did not explicitly test stability across entirely different state accountability systems or conduct external validation on a held-out cohort of districts, leaving generalizability to non-U.S. or post-2019 contexts unexamined.

Future work will address these limitations by incorporating longitudinal funding streams, experimenting with finer-grained achievement metrics, and testing more sophisticated fusion architectures (e.g., hybrid feature-decision level with attention mechanisms) while maintaining the interpretable, policy-relevant focus that guided the present study

VI. Conclusion & Future Work

This study adapted the multi-source data fusion framework of Chango et al. [2] to determine which aggregation of school district funding per-pupil, aggregate sum, or percentage best predicts district-level academic achievement tertiles when fused with standardized test scores and socioeconomic covariates. Using national data from SEDA 6.0 and NCES F-33 (2018–19), a final analytic sample of 12,100 U.S. school districts was constructed and four progressive fusion strategies were evaluated with six white-box classifiers on both numerical and discretized data [8].

Results showed that the simplest naïve-merge approach (Experiment 1) consistently yielded the strongest models. The aggregate-sum formulation of finance data achieved the highest performance (68.94 % accuracy and macro-averaged

AUC of 0.8538 on discretized data). In contrast to Chango et al. [2], feature selection and hybrid ensembles slightly reduced predictive power, confirming that socioeconomic variables (especially free/reduced-price lunch eligibility and the SES composite index) overwhelmingly dominate classification. Finance features contributed meaningfully only when retained in raw-sum form, with local revenue per pupil emerging as the primary mediator between neighborhood wealth and achievement. By satisfying Meng et al.'s [3] criteria for efficiency, benefit, robustness, and extensibility, this work extends the Chango framework from individual students to the district-finance domain and offers a clear methodological recommendation: preserve absolute-scale funding variables rather than over-normalizing.

Future research should extend the current cross-sectional design to longitudinal fusion using multi-year panels, enabling the modeling of lagged effects and causal dynamics. Additional directions include testing finer-grained achievement metrics with more sophisticated yet interpretable classifiers (e.g., gradient-boosted ensembles with SHAP explanations), applying the framework to international or school-level data, and exploring hybrid low-level/high-level fusion architectures that explicitly weight finance streams by their correlation with local wealth. These extensions will further strengthen the generalization and policy relevance of data-fusion methods in educational finance research.

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